Jun Young (Jun) Park

Contact Information

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Current Position

July 2020 -	**Assistant Professor**, University of Toronto Department of Statistical Sciences (67%) Department of Psychology (33%)
June 2021-	**Affiliate Scientist (status-only)**, The Centre for Addiction and Mental Health (CAMH)

Research Interests

Methodological:	Modeling of correlated data (multivariate time-series, spatiotemporal data, networks), Resampling-based inference (permutation and bootstrapping); Integration of high-dimensional data, Statistical learning.
Scientific:	Neuroimaging, data integration, statistical genetics and genomics

Education

May 2020	**PhD in Biostatistics**, University of Minnesota - Twin Cities Thesis: *Statistical modeling and inference for neuroimaging and genomics data* Advisor: Mark Fiecas
June 2012	**BA in Mathematics/Statistics**, Carleton College

Publications

♦ *indicates a student author.* ♣ *indicates a (one of) corresponding author(s).*

Published/accepted

1. ♦Zhang, R., Tuzhilina, E., ♣**Park, JY**. (2026+) Sparse covariate-driven factorization of high-dimensional brain connectivity with application to site effect correction. ***Biostatistics***. DOI: 10.1093/biostatistics/kxag034
2. ♦ Veitch, D., He, Y., ♣**Park, JY**. (2026). Rank-adaptive covariance testing with applications to

genomics and neuroimaging. ***Biometrics***. 82(1), ujag052. DOI: 10.1093/biomtc/ujag052
An earlier version of this manuscript won a distinguished student paper award for ENAR 2024.

3. ♦Pan, R., Weinstein, SM., Tu, D., Hu, F., Tanrıverdi, B, ♦Zhang, R., Vandekar, SN., Baller, EB., Gur, RC., Gur, RE., Alexander-Bloch, AF., Satterthwaite, TD., ♣**Park, JY**. (2025) Mapping individual differences in intermodal coupling in neurodevelopment. ***Imaging Neuroscience***. 3: IMAG.a.156. DOI: 10.1162/IMAG.a.156

4. ♦Tian, Y., Felsky, D., Gronsbell, J., ♣**Park, JY**. (2025) Leveraging multimodal neuroimaging and GWAS for identifying modality-level causal pathways to Alzheimer's disease. ***Imaging Neuroscience***. 3: imag_a_00580. DOI: 10.1162/imag_a_00580

5. ♦Zhang, R., ♦Chen, L., Oliver, LD., Voineskos, AN., ♣**Park, JY**. (2024) SAN: Mitigating spatial covariance heterogeneity in cortical thickness data from multiple sites or scanners. ***Human Brain Mapping***, 2024. 45(7), e26692 DOI: 10.1002/hbm.26692

6. ♦Pan, R., Dickie, EW., Hawco, C., Reid, N., Voineskos, AN., ♣**Park, JY**. (2024) Spatial-extent inference for testing variance components in reliability and heritability studies**.** ***Imaging Neuroscience.*** 2: imag-2-00058. DOI: 10.1162/imag_a_00058

7. ♦Zhang, R., Oliver, LD., Voineskos, AN., ♣**Park, JY**. (2023) RELIEF: a structured multivariate approach for removal of latent inter-scanner effects. ***Imaging Neuroscience***. 1: imag-1-00011 DOI: 10.1162/imag_a_00011
This manuscript won a student paper award (runner-up) for SMI 2022.

8. ♦Bouffard, NR., Golestani, A., Brunec, IK., Bellana, B., **Park, JY.**, Barense, MD., Moscovitch, M. (2023) Single voxel autocorrelation uncovers gradients of temporal dynamics in the hippocampus and entorhinal cortex during rest and navigation. ***Cerebral Cortex***. 33(6): 3265-3283. DOI: 10.1093/cercor/bhac480

9. ♦Weinstein, SM., Vandekar, SN., Baller, EB., ♦ Tu, D., Adebimpe, A., Tapera, TM., Gur, RC., Gur, RE., Detre, J., Raznahan, A., Alexander-Bloch, AF., Satterthwaite, TD., Shinohara, RT., ♣**Park, JY**. Spatially-enhanced clusterwise inference for testing and localizing intermodal correspondence. ***Neuroimage***. 264, 119712. DOI: 10.1016/j.neuroimage.2022.119712

10. ♣**Park, JY.**, Fiecas, M. (2022) CLEAN: Leveraging spatial autocorrelation in neuroimaging data in clusterwise inference. ***Neuroimage***. 255, 119192. DOI:10.1016/j.neuroimage.2022.119192

11. Lock, EF., **Park, JY.**, Hoadley, HA. (2022) Bidimensional linked matrix factorization for pan-omics pan-cancer analysis. ***Annals of Applied Statistics***. 16(1): 193-215. DOI: 10.1214/21-AOAS1495

12. ♣**Park, JY.**, Fiecas, M. (2021) Permutation-based inference for spatially localized signals in longitudinal MRI data. ***Neuroimage***. 239, 118312. DOI: 10.1016/j.neuroimage.2021.118312
This manuscript won a student paper award for ASA Statistics in Imaging student paper competition 2020.

13. ♣**Park, JY**, Polzehl, J., Chatterjee, S., Brechmann, A., Fiecas, M. (2020) Semiparametric modeling of time-varying activation and connectivity in task-based fMRI data. ***Computational***

Statistics & Data Analysis. 150, 107006. DOI: 10.1016/j.csda.2020.107006
This manuscript won a student paper award for SMI 2019 and ASA Statistics in Imaging student paper competition 2019.

14. **Park, JY**, Lock, EF. (2020) Integrative factorization of bidimensionally linked matrices. ***Biometrics***. 76(1):61-74. DOI: 10.1111/biom.13141

15. Wu, C., **Park, JY**., Guan, W., Pan, W. (2018) An adaptive gene-based test for methylation data. ***BMC Proceedings***. 12(Supp 1):68. DOI: 10.1186/s12919-018-0126-9

16. **Park, JY.,** Wu, C., Pan, W. (2018) An adaptive gene-level association test for pedigree data. ***BMC Genetics***. 19(Supp 1):68. DOI: 10.1186/s12863-018-0639-2

17. **Park, JY.**, Wu, C., Basu, S., McGue, M., Pan, W. (2018) Adaptive SNP-set association testing in generalized linear mixed models with application to family studies. ***Behavior Genetics***. 48(1):55-66. DOI: 10.1007/s10519-017-9883-x

Submitted/under review

18. St.Clair, K., **Park**, **JY.**, Gray, BR., Capers, RS. (2026+) Modeling occupancy probabilities hierarchically, given misclassification and spatial dependence.
Status: Submitted.

19. ♦Chen, Y., Rajji, TK., Zhukovsky, P., …, ♦Pan, R., **Park, JY.**, Pollock, BG., Mulsant, BH., Voineskos, A. (2026+) Impact of Cognitive Remediation Combined with Transcranial Direct Current Stimulation on Gray Matter Macrostructure and Microstructure in Older Adults at Risk for Alzheimer’s Dementia.
Status: Under minor revision.

20. ♦Pan, R., He, Y., ♣**Park, JY**. (2026+) Boosting multi-view association testing via devariation. DOI: 10.48550/arXiv.2603.26981
Status: In revision.
An earlier version of this manuscript won a student paper award (runner-up) for ASA Statistics in Imaging section 2025.

21. ♣**Park, JY**., MacDonald, PW. (2026+) Group-regularized matrix factorization for fast and reliable module discovery in pan-omics pan-cancer studies. DOI: 10.48550/arXiv.2608.04826
Status: Submitted.

In preparation

22. ♦Wang, L., ♦Yang, X., ♣**Park, JY**. Integrating binning with inverse probability of censoring weighting for improved risk prediction with machine learning.

Grants and Supports

2026-2031 **Collaborative Research and Training Experience (CREATE) program,** Natural Sciences and Engineering Research Council of Canada, *Collaborator*, “CANSSI Genomic Data Science CREATE”

2023-2025 **Labatt Family Innovation Fund**, University of Toronto, *Co-Investigator* (PIs: Drs.

Colin Hawco, Ishrat Husain, Joshua Rosenblat), “Evaluating psilocybin assisted psychotherapy in depression using neuroimaging”

2023-2025 **Connaught New Researcher Award**, University of Toronto, *Principal Investigator*, “Fostering open science and reproducibility in neuroimaging studies by leveraging summary statistics” ($20,000 CAD)

2023-2025 **Accelerator Grant**, University of Toronto McLaughlin Centre, *Lead Principal Investigator* (Co-PI: Drs. Daniel Felsky, Jessica Gronsbell), “Leveraging multi-modal neuroimaging for the discovery of modality-specific genetic interactions for Alzheimer’s disease” ($75,000 CAD)

2022-2027 **Discovery Grant**, Natural Sciences and Engineering Research Council of Canada, *Principal Investigator*, “Spatial-extent inference and prediction in brain imaging data” ($95,000 CAD)

2022-2027 **Discovery Launch Supplement**, Natural Sciences and Engineering Research Council of Canada, *Principal Investigator*, “Spatial-extent inference and prediction in brain imaging data” ($12,500 CAD)

2022-2025 **Catalyst Grant**, University of Toronto Data Science Institute, *Nominated Principal investigator* (Co-PI: Laurent Briollais, Michael Wilson) “Removing unwanted variations from heterogeneous neuroimaging and genomic data” ($100,000 CAD)

2022-2027 **Multidisciplinary Doctoral Program**, CANSSI Ontario, *Supervisor* ($50,000 CAD equivalent)

2021-2025 **Insight Grant**, Social Sciences and Humanities Research Council, *Collaborator*, (PI: Felix Cheung) “Revisiting the income-happiness paradox: testing the rapidity of income growth as a key to happiness”

Awards & Honors

2026 **Resource Allocation Competition**, Digital Research Alliance of Canada

2023 **Connaught New Researcher Award**, The Connaught Fund

2023 **Resource Allocation Competition**, Digital Research Alliance of Canada

2020 **Student Paper Award (runner-up)**, American Statistical Association (Statistics in Imaging section)

2019 **Student Paper Award (runner-up)**, American Statistical Association (Statistics in Imaging section)

2019 **Student Award**, Statistical Methods in Imaging (SMI) conference

2019 **Biostatistics Best Student Paper Award**, University of Minnesota Division of Biostatistics

2019 **MnDRIVE PhD Informatics Fellowship**, University of Minnesota

2014 **Outstanding Teaching Assistant Award,** University of Minnesota Division of

Biostatistics

2013 **Dean's PhD Scholar's Award**, University of Minnesota School of Public Health

Presentations

Talks

2026 *Statistical considerations on intermodal coupling analysis*, International Chinese Statistical Association (ICSA) Canada chapter meeting, invited session

2026 *Sparse covariate-driven factorization of high-dimensional brain connectivity with application to site effect correction*, Joint Statistical Meetings, invited session

2026 *Preparing good data for more reproducible science in multi-site neuroimaging studies*, Miami University, Department of Statistics

2026 *Statistical considerations on intermodal coupling analysis*, Statistical Methods in Imaging (SMI) conference, invited session

2025 *Sparse covariate-driven factorization of high-dimensional brain connectivity with application to site effect correction,* Korean Statistical Society Winter Conference

2025 *Preparing good data for more reproducible science in multi-site neuroimaging studies* University of Waterloo Department of Statistics and Actuarial Science

2025 *Mapping individual differences in intermodal coupling in neurodevelopment*, Joint Statistical Meetings (JSM), invited session

2025 *Boosting multi-view association testing via devariation*, 2025 Statistical Society of Canada (SSC) Annual Meeting, invited session

2025 *Boosting multi-view association testing via devariation*, Statistical Methods in Imaging (SMI) conference, invited session

2025 *Boosting multi-view association testing via devariation*, Seoul National University Department of Statistics

2025 *Leveraging multimodal neuroimaging and GWAS for identifying modality-level causal pathways to Alzheimer's disease.* Eastern North American Region (ENAR) meeting, invited session

2024 *Promises of covariance harmonization in multi-site neuroimaging studies*, Computational and Methodological Statistics (CMStatistics), invited session

2024 *Brain Behavior and Synchronization*, University of Chicago Institute for Mathematical and Statistical Innovation, invited panel

2024 *Boosting multi-view association testing via devariation,* The 7th International Conference on Econometrics and Statistics, invited session

2024 *Spatial-extent inference in neuroimaging studies*, Korean Statistical Society Summer Conference, invited session

2024 *Leveraging multimodal neuroimaging and GWAS for identifying modality-level causal* *pathways to Alzheimer's disease*, Statistical Methods in Imaging (SMI) conference, invited session

2024 *Spatial-extent inference in neuroimaging studies,* New England Statistics Symposium invited session

2023 *Mitigating inter-scanner biases in high-dimensional neuroimaging data via spatial Gaussian process*, Computational and Methodological Statistics (CMStatistics), invited session

2023 *Spatial-extent inference in neuroimaging studies,* University of California Santa Cruz, Department of Statistics

2023 *Spatially-enhanced clusterwise inference for testing and localizing intermodal correspondence*, Joint Statistical Meetings (JSM), invited session

2023 *Fast and powerful spatial-extent inference for testing variance components in reliability and heritability studies*, The 6th International Conference on Econometrics and Statistics, invited session

2023 *Mitigating spatial covariance heterogeneity in cortical thickness data collected from multiple scanners or sites*, NeuroImaging Statistics satellite meeting to the 2023 Organization for Human Brain Mapping

2023 *Spatially-enhanced clusterwise inference for testing and localizing intermodal correspondence*, Statistical Methods in Imaging (SMI) conference, invited session

2023 *Spatial-extent inference for testing variance components in reliability and heritability studies*, Banff International Research Station (BIRS) workshop at Casa Matemática Oaxaca, Mexico, invited session

2023 *Spatially-enhanced clusterwise inference for testing and localizing intermodal correspondence*, Eastern North American Region (ENAR) meeting, invited session

2022 *CLEAN: Leveraging spatial autocorrelation in neuroimaging data in clusterwise inference*, Computational and Methodological Statistics (CMStatistics), invited session

2022 *CLEAN: Leveraging spatial autocorrelation in neuroimaging data in clusterwise inference,* Joint Statistical Meetings (JSM), invited session

2022 *A structured multivariate approach for removal of latent batch effects*, University of Toronto Data Science Institute

2022 *A structured multivariate approach for removal of latent batch effects,* Eastern North American Region (ENAR) meeting

2021 *A structured multivariate approach for removal of latent batch effects*, PennSIVE Center, University of Pennsylvania Perelman School of Medicine

2021 *Permutation-based spatial scanning for detection of fMRI activation clusters*, Eastern North American Region (ENAR) meeting

2020 *Permutation-based inference for spatially localized signals in longitudinal MRI data*, Joint Statistical Meeting (JSM)

2020 *Permutation-based inference for spatially localized signals in longitudinal MRI data,* Eastern North American Region (ENAR) meeting

2020 *Permutation-based inference for spatially localized signals in longitudinal MRI data,* Wake Forest University School of Medicine

2020 *Permutation-based inference for spatially localized signals in longitudinal MRI data,* Vanderbilt University Medical Center

2020 *Permutation-based inference for spatially localized signals in longitudinal MRI data,* Columbia University

2019 *Integrative factorization of bidimensionally linked matrices*, International Chinese Statistical Association (ICSA) Applied Statistics Symposium

2019 *Semiparametric modeling of time-varying activation and connectivity in task- based fMRI data*, Joint Statistical Meeting (JSM)

2019 *Semiparametric modeling of time-varying activation and connectivity in task- based fMRI data*, Statistical Methods in Imaging (SMI)

2019 Semiparametric modeling of time-varying activation and connectivity in task- based fMRI data, Eastern North American Region (ENAR) meeting

2018 *Adaptive SNP-set association testing in generalized linear mixed models with application to family studies*, Eastern North American Region (ENAR) meeting

Posters

2026 *Sparse covariate-driven factorization of high-dimensional brain connectivity for site effect correction*, The Organization of Human Brain Mapping (OHBM) meeting

2026 *Mapping individual differences in intermodal coupling in neurodevelopment*, The Organization of Human Brain Mapping (OHBM) meeting

2024 *SAN: Mitigating spatial covariance heterogeneity in cortical thickness data collected from multiple scanners or sites*, The Organization of Human Brain Mapping (OHBM) meeting

2024 *RELIEF: a structured multivariate approach for removal of latent inter-scanner effects*, The Organization of Human Brain Mapping (OHBM) meeting

2023 *Spatial-extent inference for testing variance components in reliability and heritability studies*, The Organization of Human Brain Mapping (OHBM) meeting

2022 *A structured multivariate approach for removal of latent batch effects*, The Organization of Human Brain Mapping (OHBM) meeting

2021 *SpLoc: Permutation-based spatial scanning for detecting fMRI group-level activation clusters*, Statistical Methods in Imaging (SMI) conference

2019 *Integrative factorization of bidimensionally linked matrices,* Twin Cities ASA Chapter Meeting

2019 *Integrative factorization of bidimensionally linked matrices*, UMN School of Public Health Research Day

2017 *Adaptive SNP-set association testing in generalized linear mixed models with application* *to family studies*, UMN Minnesota Supercomputing Institute (MSI) Research Exhibition

2017 *Adaptive SNP-set association testing in generalized linear mixed models with application to family studies*, UMN School of Public Health (SPH) Research Day

Teaching

Course instructor (University of Toronto)

PSY 305: Treatment of psychological data (undergraduate level)
Offered: Winter 2023, 2024, 2025, 2026, Fall 2026 (scheduled)

STA442 Methods of applied statistics (undergraduate level)
Offered: Fall 2022, 2023, 2024, 2025

STA447/2006 Stochastic processes (undergraduate and graduate level)
Offered: Winter 2021, 2022

STA1008 Applied statistics (graduate level)
Offered: Fall 2020, 2021, 2022, 2023

STA2101 Methods of Applied Statistics I (graduate level)
Offered: Fall 2024, 2025, 2026 (scheduled)

Reading course supervisor (University of Toronto)

STA4000 Supervised Reading Project I (graduate level)
Offered:
Fall 2025 (with Evan Shamov)
Fall 2025 (with Ruilin Bai)
Fall 2025 (with Amanda Wai Yu Ng)
Fall 2022 (with Ruyi Pan)

STA496 Readings in Statistics (undergraduate level)
Offered:
Fall 2024 - Winter 2025 (with Ian Yiyang Zhang)
Summer 2023 (with Haonan Gao)
Fall 2020 - Winter 2021 (with Joanna Lo)

Teaching assistant (University of Minnesota)

PUBH 6141: Biostatistical literacy (MPH level)

PUBH 6450: Biostatistics I (MPH level)

PUBH 6461: Exploring and visualizing data in R (MPH level)

PUBH 7420: Clinical trials (MS level)

PUBH 7430: Statistical methods for correlated data (MS level)

PUBH 8401: Linear models (PhD level)

PUBH 8475: Statistical learning and data mining (PhD level)

Services

Service to the profession:

Conferences

Session organizer, JSM 2026 Invited session
Session organizer, JSM 2025 Invited session
Session organizer, JSM 2024 Topic-contributed session
Session organizer, SMI 2024 Invited session
Session organizer, SMI 2023 Invited session
Session organizer, JSM 2023 Topic-contributed session
Session organizer, ENAR 2023 Invited session
Session organizer, ENAR 2022 Invited session
Session chair, ICSA Applied Statistics Symposium 2019
Session chair, ENAR 2019

Journal review

Annals of Applied Statistics
Bioinformatics
Biometrics
Biometrika
Computational Statistics & Data Analysis
Frontiers in Neuroscience
Human Brain Mapping
Imaging Neuroscience
Journal of American Statistical Association
NeuroImage
Statistics in Biosciences
Statistics in Medicine
WIREs Computational Statistics

Conference review

Reviewer of the student paper competition, ASA Statistics in Imaging section 2022, 2023, 2024
Reviewer of the student paper competition, Statistics in Imaging conference, 2023, 2024

Service to the university/department

Member of various university and department-level committees since 2020.

Students (at the University of Toronto)

PhD students (Doctoral thesis supervisor)

Rongqian Zhang
Program: PhD in Statistics

Period: 2021-2025 (completed)
Thesis: Novel statistical methods for mitigating inter-site variability in neuroimaging studies
Placement: Postdoctoral scholar, *University of California San Francisco*

David Veitch (co-suprevised by Dr. Zhou Zhou)
Program: PhD in Statistics
Period: 2022-2025 (completed)
Thesis: Two-sample testing and changepoint detection for high-dimensional data
Placement: Systematic Macro Research Analyst, *Connor, Clark & Lunn*

Yuan Tian (co-supervised by Dr. Jessica Gronsbell)
Program: PhD in Statistics
Period: 2021-present

Ruyi Pan (co-supervised by Drs. Nancy Reid and Aristotle Voineskos)
Program: PhD in Statistics
Period: 2022-present

William Groff (co-supervised by Dr. Elena Tuzhilina)
Program: PhD in Statistics
Period: 2025-present

Ruilin Bai
Program: PhD in Statistics
Period: 2025-present

Junjie Ma (co-supervised by Nancy Reid)
Program: PhD in Statistics
Period: 2026-present

PhD students (Examiner for the oral exam)

Ashley Moo-Choy (in progress)
Program: PhD in Psychiatric genetics

Junhao Zhu (completed in 2025)
Program: PhD in Statistics

Ziang Zhang (completed in 2024)
Program: PhD in Statistics

Fan Wang (completed in 2022)
Program: PhD in Statistics

Lin Zhang (completed in 2021)
Program: PhD in Statistics

Wei Q. Deng (completed in 2021)
Program: PhD in Statistics

PhD students (STAGE program mentor)

Yuan Tian

Program: PhD in Statistics
Period: 2022-present

Tara Henechowicz

Program: PhD in Neuroscience
Period:2022-2025

Master students (Research supervisor)

Hainan Xu

Period: 2023-2024
Placement: Associate Data Scientist, *Co-operators*

Master students (Supervisory committee member)

Cathlyn (Yihan) Chen

Program: Master of Science (MSc) in Medical Science (IMS)

Undergraduate students (Research supervisor)

Yinghao Li (visiting research student from Tsinghua University)

Period: Summer 2025
Placement: Master of Science in Statistics, *University of Chicago*
Note: Supported by the *Mitacs Globalink Research Internship*

Liyan Wang

Period: Fall 2024 - Summer 2025
Placement: Master of Quantitative Finance, *University of Waterloo*
Note: Partially supported by the *University of Toronto Data Science Institute*

Alice Zhang

Period: Summer 2024
Placement: Master of Arts in Statistics, *University of California Berkeley*
Note: Supported by the *University of Toronto Research Excellence Award (UTEA)*

Yun Zhu (Helena) Li

Period: Summer 2024
Placement: Master of Science in Health Data Science, *Harvard University*
Note: Supported by the *University of Toronto Research Excellence Award (UTEA)*

Zhengdan Li

Period: Summer 2022
Placement: Master of Science in Statistics & Data Science, *Stanford University*
Note: Supported by the *University of Toronto Research Excellence Award (UTEA)*

Linxi Chen

Period: Summer 2022

Placement: Master of Science in Statistical Science, *University of Oxford*
Note: Supported by the *University of Toronto Data Science Institute, Summer Undergraduate Data Science (SUDS) Opportunities Program*

Xiaoli Yang
Period: 2021
Placement: Master of Science in Statistics & Data Science, *Stanford University*

Miscellaneous

Citizenship	Republic of Korea (South Korea)
Languages	English, Korean
Membership	American Statistical Association (ASA), Statistical Society of Canada (SSC), Organization of Human Brain Mapping (OHBM), Korean International Statistics Society (KISS)